



Accuracy of Apple Watch to Measure Cardiovascular Indices in Patients with Cardiac Diseases: Observational Study

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ABSTRACT

The validity of Apple Watch for measuring heart rate (HR) and oxygen saturation (SpO₂) in patients with cardiac diseases is still unclear. Therefore, this study aimed to investigate the accuracy of the Apple Watch in measuring HR and SpO₂ in patients with cardiac diseases. A cross-sectional study recruited 260 cardiac patients, including 190 with regular heart rhythm and 70 with cardiac arrhythmia. Each patient wore the Apple Watch alongside a Polar HR monitor at rest, during and after mild- to moderate-intensity exercise sessions, and wore the Apple Watch alongside a Contec pulse oximeter at rest and after exercise. The Apple Watch showed excellent validity (ICC = 0.100) in measuring the HR at rest, during mild- to moderate-intensity exercise, and after exercise in cardiac patients, as well as in measuring SpO₂ at rest (ICC = 0.100) and after exercise (ICC = 0.92). However, the validity of the Apple Watch for measuring SpO₂ decreased slightly after exercise (ICC = 0.85; good validity), especially in patients with an irregular heart rhythm. Overall, the Apple Watch appears valid for measuring HR and SpO₂ at rest and after exercise, and for measuring HR during mild- to moderate-intensity training in cardiac patients.

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Measuring the heart rate (HR) is important to estimate the suitable exercise training intensity and to determine the recommended exercise levels (1). Exercise is one of the important elements to treat cardiac disease, which is a leading cause of death worldwide (2–4). The 12-lead electrocardiogram (ECG) is the gold standard to measure HR, but it may not be suitable for use outside the laboratory; therefore, using the Polar chest strap is a practical alternative to be used outside the lab, and it has been tested against the 12-lead ECG during exercise and at rest in both healthy subjects (5–7) and cardiac patients (8). The HR following exercise is called HR recovery and it is important to predict mortality risk and cardiorespiratory fitness (9).

There is a linear relationship between HR and oxygen consumption (10). Monitoring oxygen saturation (SpO₂) is recommended for patients with hypoxemia to help extend their survival (11). Some cardiac patients have hypoxemia, particularly after cardiac arrest (12–14). A pulse oximeter is the gold standard for measuring SpO₂ (15). The Contec CMS50DL pulse oximeter meets the International Organization for Standardization (ISO) criteria for validity (16) and is approved by the Food and Drug Administration (FDA) (17).

All versions of the Apple Watch can measure HR, but the latest versions (Series 6 and 7) can also measure SpO₂. Photoplethysmography (PPG) is a non-invasive measurement technique used to evaluate HR via a wrist-worn device by detecting changes in blood flow and its associated color changes with each heartbeat. The PPG has been used to measure SpO₂ and blood pressure in various medical devices (18). PPG technology may also help in the early detection of some cardiovascular diseases such as atherosclerosis and atrial fibrillation (AF) (19).

The new feature of measuring the blood SpO₂ is available only in the Apple Watch Series 6 and 7. To date, only one study has examined the validity of the Apple Watch compared to two pulse oximeters for measure SpO₂ at rest and HR during exercise ($r = 0.995$, $P < 0.001$) in 100 participants, including 16 healthy subjects and 84 patients with medically controlled pulmonary diseases [23 patients with chronic obstructive pulmonary disease (COPD) and 61 patients with interstitial lung diseases]. They found that the Apple Watch showed a strong positive correlation with the pulse oximeter for SpO₂ ($r = 0.81$, $P < 0.0001$) (20). Moreover, the study suggested testing the validity of the Apple Watch for measuring blood SpO₂ in other groups of patients. A systematic review of recent studies investigating the accuracy of the Apple Watch in measuring SpO₂ in children and adults found that the Apple Watch has no strong systematic bias. However, no study has specifically investigated the accuracy of the Apple Watch in measuring SpO₂ in cardiac patients with sinus rhythm and irregular HR at rest and after exercise, or the impact of skin color on measuring SpO₂. The review suggests that further research is needed in this area (21). Spaccarotella et al. measured SpO₂ and HR ($r = 0.98$, $P < 0.0001$) during exercise in cardiac patients with sinus rhythm, but there were no cardiac patients with irregular HR, and they did not measure the accuracy of the Apple Watch in measuring SpO₂ after exercise, or the impact of skin color while measuring SpO₂ (22). Another recent study measured HR during mild- to moderate-intensity exercises and SpO₂ at rest in patients with chronic diseases such as diabetes, hypertension, and dyslipidemia. They found that the Apple Watch was accurate to measure HR during exercise ($r = 0.99$, $P < 0.001$) and SpO₂ at rest ($r = 0.92$, $P < 0.001$) in patients with chronic diseases and recommends validating these findings with other patient groups, such as cardiac patients (23). The rationale for evaluating the Apple Watch in cardiac patients is that they need to measure HR and SpO₂ to assess and quantify cardiovascular risk and to monitor cardiovascular responses in day-to-day clinical cardiac practice (22). Therefore, this study aimed to evaluate the accuracy of the Apple Watch for measuring HR at rest, during, and after exercises, and to measure SpO₂ at rest and after exercises in patients with cardiac diseases with both normal and abnormal HR patterns.

METHODS

A cross-sectional study recruited 260 male patients with cardiac diseases (mean [SD]; age 47 (15) years), including 190 ischemic patients with regular heart rhythm and 70 patients with irregular rhythm, and they all completed the study. Out of 260 patients, 160 patients had white skin, 100 had brown skin, and no patient had black skin. This study was approved by the biomedical research ethics committee at Umm Al-Qura University (HAPO-O2-K-012-2022-01-910). The

study followed the principles of the Declaration of Helsinki of 1975, revised in 2000, and was conducted at the Umm Al-Qura University Medical Center. This study was registered at ClinicalTrials.gov (NCT05199844). Each patient signed a written consent approving their volunteering to participate in this study and agreeing to the publication of the results.

DEVICES AND DATA COLLECTION

After medical counseling and physician referral, each participant wore a Polar chest strap (H10, Polar Electro, OY, Finland) and an Apple Watch (Series 8, watchOS 9.0, Apple Inc., California, USA) on the left wrist, which was connected to an iPhone (iPhone 11, iPhoneOS 16.1, Apple Inc., California, USA) (all patients rested their hands on the plastic handrails of the cycle) at rest for 5 min, during exercise for 16 min, and after exercise for 3 min. The exercise session was for 16 min in duration at mild- to moderate-intensity exercise [40%–70% of heart rate reserve (HRR)] according to the patient's ability using a cycle ergometer (Longstyle, China). The patient wore a Contec pulse oximeter (CMS50DL, Contec Medical Systems Co., Ltd., China) and an Apple Watch at rest and after exercise to measure SpO₂.

The HR was recorded every 30 s during the whole period and the oxygen saturation was recorded five times at rest and after exercise simultaneously from the Apple Watch and the Polar chest strap and from the Apple Watch and the Contec pulse oximeter, respectively, to obtain the data.

The exercise intensity during the session was calculated based on the HRR. The HRR was calculated as: (maximum HR – resting HR) + resting HR. The maximum HR was calculated based on the following equations: for cardiac patients with no β -blocker, maximum HR = 206.9 – (0.67 × age) (24), and for cardiac patients with β -blockers, maximum HR = 164 – (0.7 × age) (25).

Inclusion criteria: cardiac patients with myocardial infarction (6 weeks post-insult), coronary bypass graft surgery (CABG), valve diseases, stable AF, heart failure (HF)-I, II, and III based on the New York Heart Association classification of heart failure (26), and a resting ejection fraction (EF) >50%.

Exclusion criteria: cardiac patients with unstable angina, uncontrolled high blood pressure, unstable arrhythmia, the presence of complex ventricular arrhythmias, ST-segment depression ≥ 2 mm from baseline during exercise testing or recovery, and pacemaker patients.

SAMPLE SIZE

The suitable number of participants was calculated through the online G-Power program (https://download.cnet.com/GPower/3000-2054_4-10647044.html), considering the alpha error probability = 0.05, power = 0.95, and Cohen's f 'effect size' = 0.23 (small effect size) (27), resulting in a total sample size of 248 participants to provide reliable results. An additional 12 participants were added to compensate for any drop or withdrawal. Additionally, the sample size required to clarify significant differences in the present study was chosen considering the guidelines of the past studies in the same field (28–35).

DATA ANALYSIS

All data were analyzed using SPSS version 29 (IBM Corp, Chicago, IL, USA). The data are presented as the mean and standard deviation. The Shapiro–Wilk test was used to verify the normal distribution of the data. The mean difference (MD) and standard deviation of the mean difference (SDD) were calculated to construct Bland–Altman plots. Bland–Altman plots were used to test the bias (MD) and the limits of agreement (LoA, MD $\pm$ 1.96 * SDD) of the data. Intraclass correlation coefficients (ICCs) were used to determine the correlation between the Apple Watch and Polar chest strap for the HR variable and between the Apple Watch and Contec pulse oximeter for the SpO₂ variable. The strength of the ICC was interpreted based on the Fokkema et al. suggestion as follows: ICC >0.90 was excellent, 0.75 to 0.90 was good, 0.60 to 0.75 was moderate, and <0.60 was low (28). An independent t -test was performed to calculate the difference between white and brown skin color patients for the SpO₂ variable at rest. The P value was set at 0.05 for significant results.

RESULTS

HEART RATE

There was an excellent correlation for HR measurements in patients with regular and irregular rhythms and for overall cardiac patients (Table 1). The LoA were (upper, lower LoA) 0.28, -0.26, and the bias (mean difference) was 0.01 for HR all in cardiac patients with no outlier (Figure 1).

OUTCOMES	ICC (95% CI)	P VALUE	MEAN (SD) OF PRACTICAL (APPLE WATCH)	MEAN (SD) OF CRITERION (POLAR)
Heart rate				
Cardiac patients with regular rhythm	0.100 (0.100–0.100)	<0.001	102.00 (8.06)	101.97 (8.03)
Cardiac patients with irregular rhythm	0.100 (0.100–0.100)	<0.001	106.67 (9.70)	106.63 (9.73)
All cardiac patients	0.100 (0.100–0.100)	<0.001	103.25 (8.77)	103.24 (8.75)

			MEAN (SD) OF PRACTICAL (APPLE WATCH)	MEAN (SD) OF CRITERION (CONTEC)
SpO ₂ at rest				
Cardiac patients with regular rhythm	0.98 (0.94–0.99)	<0.001	97.04 (1.09)	96.91 (0.99)
Cardiac patients with irregular rhythm	0.99 (0.93–0.100)	<0.001	96.86 (0.84)	96.80 (0.73)
All cardiac patients	0.98 (0.95–0.99)	<0.001	96.99 (1.08)	96.88 (0.91)

SpO ₂ after exercise				
Cardiac patients with regular rhythm	0.94 (0.79–0.97)	0.02	97.54 (0.84)	97.33 (0.75)
Cardiac patients with irregular rhythm	0.85 (0.15–0.97)	<0.001	97.14 (0.78)	97.03 (0.50)
All cardiac patients	0.92 (0.80–0.97)	<0.001	97.27 (0.69)	97.43 (0.83)

Table 1 Intraclass correlation for Apple Watch in cardiac patients with regular and irregular rhythm in HR and oxygen saturation at rest and after exercise. CI, confidence interval; ICC, intraclass correlation coefficient.

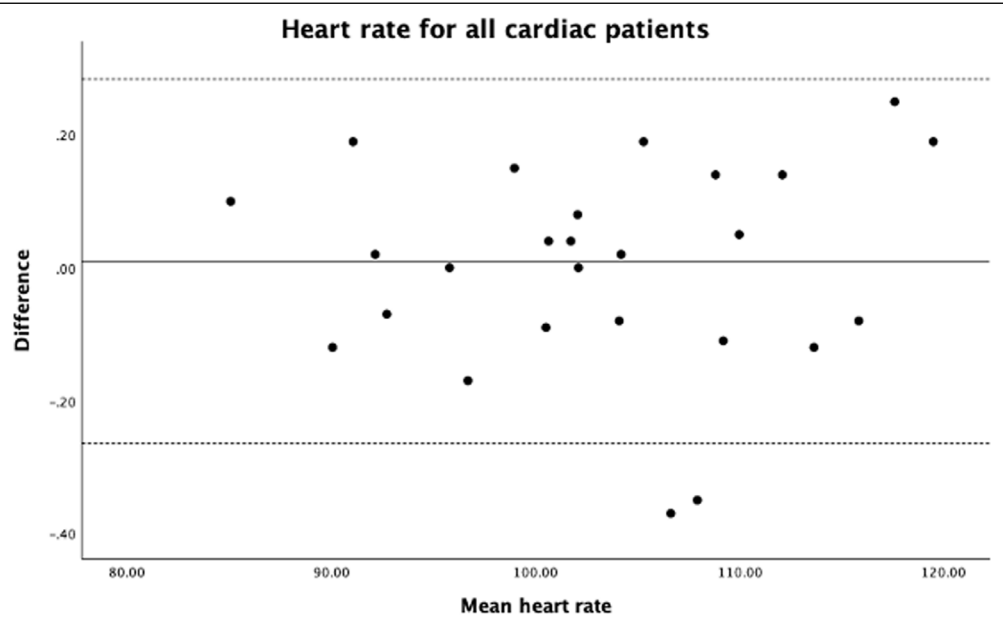


Figure 1 Heart rate for all cardiac patients.

OXYGEN SATURATION (SPO₂) AT REST AND AFTER EXERCISE

Oxygen saturation had an excellent correlation with all cardiac patients, including those with regular and irregular heart rhythms at rest (Table 1). The LoA was 0.63, -0.25. The mean difference was 0.12 for all cardiac patients at rest with an outlier of <4% SpO₂ (Figure 2). Oxygen saturation after exercise had an excellent correlation with all cardiac patients and in patients with regular heart rhythm, but there is a slight reduction in patients with irregular heart rhythms, indicating good correlation (Table 1). Overall, the correlation for SpO₂ was less after exercise than at rest in all cardiac patients (Table 1). The LoA after exercise was 0.86, -0.50, and bias (mean difference) was 0.16 with an outlier of <4% SpO₂ (Figure 3). There was no statistically significant difference between white and brown skin color patients in SpO₂ at rest ($P = 0.498$).

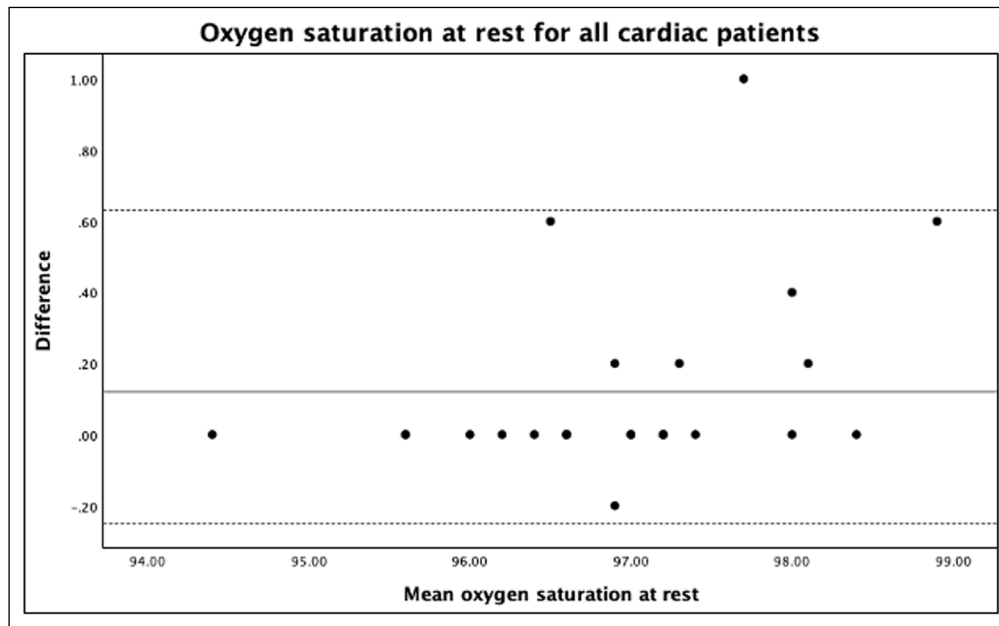


Figure 2 Oxygen saturation of all cardiac patients at rest.

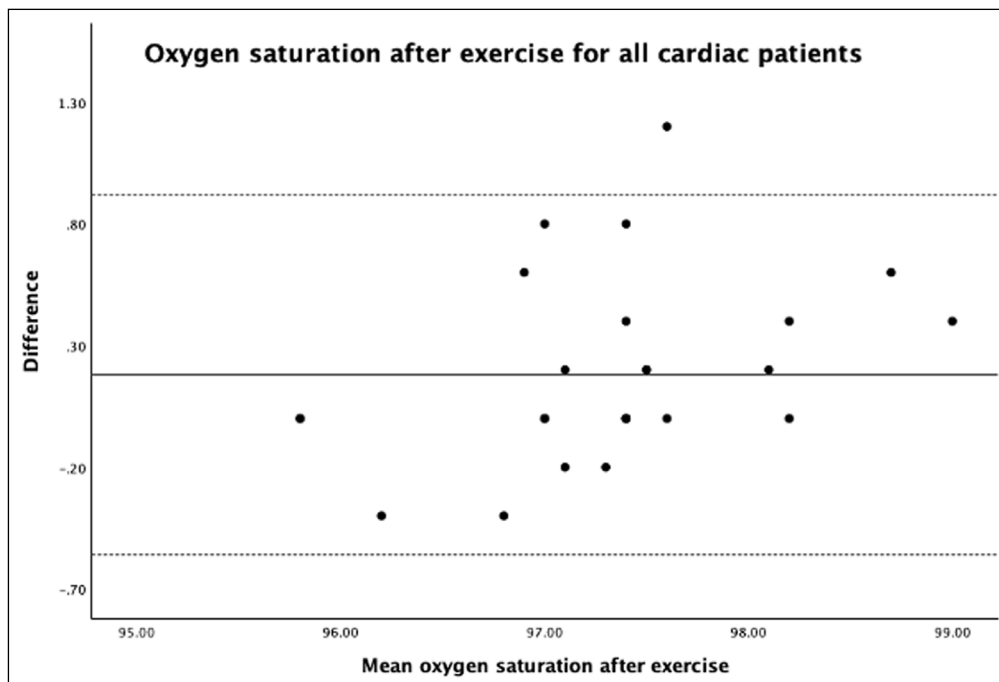


Figure 3 Oxygen saturation of all cardiac patients after exercise.

DISCUSSION

Mobile technology has grown tremendously in the last decade, with modern healthcare is shifting its focus to incorporate mobile health technology. The ability to measure HR using wrist-worn devices facilitates the continuous monitoring of patient status and cardiovascular

responses (28). Wearable technologies can be used for detecting and diagnosing health problems in patients with cardiac diseases and may help to manage and improve patient care outcomes (36). Such technology is acceptable to measure HR in daily clinical practice and to detect cardiac diseases such as irregular heart rhythms (37).

Interest in validating the accuracy of mobile health technology in the modern health care of patients with cardiac diseases has tremendously increased in the past few years. In this study, the HR and oxygen saturation were evaluated using the Apple Watch compared to gold-standard devices in 260 male cardiac patients at rest, during, and after mild- to moderate-intensity exercise sessions. For HR, the results showed excellent correlations in cardiac patients with both regular and irregular rhythms throughout the period of the exercise session, including rest, during, and after exercise. For oxygen saturation, the Apple Watch showed excellent correlation at rest for all cardiac patients with regular and irregular rhythms, with <4% outliers, and after exercise sessions in cardiac patients with regular rhythm, whereas a slightly lower (good) correlation was observed after exercise sessions in cardiac patients with irregular rhythm.

Precise monitoring of the HR is important to prescribe exercise intensity for patients with cardiac disease (38). The results of the current study indicate that the Apple Watch can be a convenient and satisfactory tool to evaluate HR in patients with cardiac disease at rest and during mild- to moderate-intensity exercise. These findings align with earlier studies that reported promising results regarding the accuracy of the Apple Watch in monitoring cardiovascular metrics in cardiac patients during rest and training. This study provides future directions to confirm these findings and to recommend the Apple Watch as an accurate monitoring device in cardiac rehabilitation settings (39).

Our results also support the use of the Apple Watch in daily clinical practice (37). Although the accuracy of using the wrist-worn Apple Watch in monitoring the HR was previously evaluated in healthy subjects (40, 28, 41–44, 30, 29), there is still an ongoing urgent need to emphasize its validity and accuracy in clinical practice (39, 28, 45).

Because of the increasing tendency of cardiac patients to rely on wrist-worn devices to monitor their HR and guide safe training intensity during cardiac rehabilitation, appropriate validation and accurate evaluation of wrist-worn devices use in the cardiac rehabilitation field is imperative (41) to stay within the recommended safe activity level and HR range during training sessions.

Regarding HR, the results of the current study were in accordance with those of Nelson et al., who reported that the accuracy of using the Apple Watch in monitoring the 24-h HR period is useful for monitoring cardiac activities in clinical settings (46). Prior research findings reported acceptable error rates when monitoring HR using the Apple Watch under controlled conditions (24). A study of 50 cardiac patients (half with AF and the other half with normal rhythm) measured HR three times a day using the Apple Watch and telemetry during rest for 2 days. The results found that the HR accuracy in AF patients ($r = 0.86$) was higher than in cardiac patients who had normal rhythm ($r = 0.64$); the overall accuracy was 0.70. However, more studies with large sample sizes are needed to confirm these findings (47). Moreover, the accuracy of the Apple Watch in measuring HR during exercise is still unknown for cardiac patients with normal rhythm and AF.

A recent study measured the HR in 80 cardiac patients, including some patients with AF, in phase II or III of cardiac rehabilitation (exercises including steady cycling and treadmill). This study compared HR from the limb leads of the ECG with the Polar HR, Apple Watch, Fitbit Blaze, Garmin Forerunner 235, and TomTom Spark Cardio (one device on each wrist), which means that only 40 patients were tested with the Apple Watch. They recorded HR at rest and at 3, 5, and 7 min of cycling and treadmill training. They found that the Polar chest strap ($r = 0.99$) was the most accurate during all exercises compared to the ECG, followed by the Apple Watch ($r = 0.80$), and the accuracy of the Apple Watch during cycling increased to 0.89 (8). However, the proportion of AF in 80 patients was low (only 12 AF) as they were mixed with the normal HR patients while testing, so it is not possible to know if the AF affected the accuracy of the Apple Watch and further research is needed in this area. The eight-time points measured in this study were good, but if the study measured the HR every 15 or 30 s, the results could be more meaningful. Falter et al. reported a clinically acceptable overall accuracy of the Apple

Watch in monitoring HR in patients with cardiovascular disease: moderate correlation [ICC = 0.729 ($P < 0.001$)] between the Apple Watch and the gold standard electrocardiogram at rest, good correlation [ICC = 0.828 ($P < 0.001$)] during moderate-intensity training, and excellent correlation [ICC = 0.958 ($P < 0.001$)] during high-intensity training (39).

Using an Apple Watch in the hospital is cheaper and easier than using an ECG in daily clinical practice. Wang et al. found excellent correlation coefficients ($r_c = 0.91$) when comparing the efficacy of a number of wrist-worn devices, including the Apple Watch 3 against the Polar H7 chest strap monitor in healthy subjects (41). Etiwy et al. found good correlation coefficients ($r_c = 0.81$) between wrist-worn devices, including the Apple Watch 3 and both the ECG and the Polar H7 chest strap monitor during cardiac rehabilitation (phases II and III) in patients with cardiovascular disorders (8). Seshadri et al. found good agreement ($r_c = 0.86$) between the Apple Watch and telemetry electrocardiogram in evaluating the AF patients' HR (44) that supports the current study findings.

Previous research reported greater accuracy by the Apple Watch in monitoring the HR compared to other wearable devices (28, 30, 41), with limited overall error (32, 28, 48), higher agreement with the ECG gold standard procedure (31, 38), and an acceptable error range even over an extended 24-h monitoring period (48).

Regarding SpO₂, although the Apple Watch showed consistency and concordance with medical-grade pulse oximeters during the SpO₂ evaluation in adults, more research is needed to resolve many accuracy-related concerns (21). The outliers in the previous studies were up to 15% SpO₂, but the current study found a lower percentage of outliers, which is <4% SpO₂, and the possible reason for this could be the controlled rest situation with the supported wrist in an armchair was recommended by the manufacturer while measuring SpO₂ in an Apple Watch (48).

The results of the current study showed that the Apple Watch can be considered for monitoring SpO₂ in patients with cardiac diseases both at rest and post-exercise. These results are in line with earlier results by Spaccarotella et al., who reported that the standard measuring device and the Apple Watch in monitoring SpO₂ in healthy subjects and in patients with cardiovascular disease at rest are similar (22). Also, the earlier research reports provide details about the accepted accuracy of the Apple Watch in monitoring the SpO₂ changes under normal and hypoxic conditions in healthy adults.

The difference in the current study results between the accuracy of the Apple Watch at rest and after exercise, particularly in cardiac patients with an irregular HR, could be due to the unstable HR influenced the results of SpO₂. According to the manufacturer, the patients should be in a complete rest with HR <150 beats per minute to get the accurate measurement of SpO₂ (48).

On the other hand, a study conducted by Pätz et al. failed to provide a clear results about the accuracy of the Apple Watch measuring SpO₂ due to improper technique and use of watch by the adult and child participants resulted in multiple incorrect readings (49).

The Fitzpatrick skin type scale's impact on measuring SpO₂ using the Apple Watch is still unclear (23). Fortunately, no obvious variability exists between participants in the skin color of patients with COVID, and these findings are in line with the current study results ($P = 0.498$), as this is the first study to examine the difference between white and brown skin color cardiac patients using an Apple Watch with regular and irregular heart rhythms, which limits the negative contributing factors since SpO₂ accuracy can be influenced by skin pigmentation (50). Further studies in different populations are needed to confirm this finding.

LIMITATIONS

The maximum HR was estimated in this study, but this is due to the difficulty of doing a real test, such as a stress test to obtain the real maximum HR in patients with cardiac diseases. Inclusion of the male gender only limits the broad generalizability of the current study results in clinical practice. However, there is no study showing any significant difference between male and female yet. Additionally, this study measured SpO₂ in patients with white and brown color skin tones with no representation from those with dark color skin. Further studies are needed to address these limitations and use more standardized evaluation procedures.

CONCLUSIONS

This study provides the first evidence for the accuracy of the Apple Watch in monitoring HR and SpO₂ in cardiac patients. These findings suggest the Apple Watch may be suitable in day-to-day clinical cardiology practice for monitoring cardiovascular responses and assessing symptoms and risks in cardiac patients.

The Apple Watch demonstrated acceptable accuracy in monitoring HR in cardiac patients with both regular and irregular rhythms. Supported by earlier studies, our study results recommend the use of Apple Watch for monitoring cardiovascular responses in cardiac patients.

DATA ACCESSIBILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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COMPETING INTERESTS

The authors have no competing interests to declare.

AUTHOR CONTRIBUTIONS

A.K., A.A., and M.A. contributed to the study's conception, design, material preparation, data collection, and writing the manuscript. A.K. and A.A. performed all data analyses. A.K., A.A., and M.A. provided comments and revised the manuscript. All authors reviewed and approved the manuscript.

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